TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Tara van Abkoude	Email	Tara.vanabkoude@tno.nl
Institute Name	TNO			
Address	Anna van Buerenplein 1, 2595 DA Den Haag			

2. Technology Overview

Title of Proposed Technology	Validation infrastructure for AEM water electrolysis technologies at industrial relevant conditions for entering EU market tailored to South Korean company's			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), **Hydrogen (27 Water Electrolysis Hydrogen Production),** Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	New & Renewable Energy	Hydrogen	R&D / Engineering Services	Testing/Inspection/Analysis Techniques
Concept of Proposed Technology	TNO Faraday Lab (Reference 1) proposes to expand and integrate advanced test infrastructure for Proton Anion Exchange Membrane (AEM) water electrolysis at industrially relevant conditions. The core concept is to further develop modular, safety-compliant test platforms that enable high-current-density operation (up to 4 A/cm²), elevated temperature (≤90 °C) and pressure (>30 bar), with precise control of water purity, gas handling and humidification. Main functions include single-cell (25–100 cm²) and short-stack (≤5 kW class) testing, dynamic load profiles, startup/shutdown protocols, accelerated stress testing (AST), and advanced electrochemical diagnostics (EIS, HFR, polarization, galvanostatic/potentiostatic modes). Potential uses span materials and component screening (membranes, catalysts, porous transport layers, bipolar plates), optimal cell integration R&D, stack engineering and degradation studies. These systems can be used by SME's for internal R&D as well as enhance Shared Research Programs in the AEM field like the SRP AEM (Reference 2)			

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Rationale for Joint R&D Project	The joint R&D leverages complementary capabilities: TNO provides electrolysis science, safety systems, and test method development, while a South Korean SME contributes cell and/or stack hardware, and manufacturing know-how for rapid iteration. The South Korean SME could also have the ambition to strengthen it's own testing capabilities for itself and/or third parties. Collaboration is required to co-develop interoperable interfaces, validate performance on representative South Korean components, cells and stacks, and prepare for certification and pilot deployment in both the EU and South Korea. The project accelerates maturation from lab-scale verification to pilot-scale prototype platforms suitable for joint demonstrations and future commercialization.
Key Words	1. AEM electrolysis 2. Cell and (short) Stack testing 3. Accelerated lifetime testing (ALT) 4. Accelerated stress testing (AST) 5. High-pressure validation

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (TRL3)	End of the Project	Level (TRL5)
R&D Objectives	• Design and build modular AEM single-cell and short-stack test benches with integrated safety and gas management. • Implement diagnostic suite (EIS 10 mHz–10 kHz, HFR, iR-correction, online gas analysis) and automated test protocols. • Develop AST protocols for catalyst/support, membrane and PTL durability under dynamic loads and transients. • Validate with partner-supplied AEM cells and 1–10 kW short-stacks under industrially relevant conditions. • Further understand degradation and cell-integration challenges through the analysis of the results of the tests.			
R&D Target Specifications	Performance targets (end of project): • Operating envelope: up to 4 A/cm² at ≤90 °C and ≤30 bar (anode/cathode), continuous; differential pressure capable. • Efficiency/overvoltage: ≤2.1 V at 2 A/cm² (AEM, 80 °C) with iR-correction; ≤2.2 V at 1 A/cm² (AEM, 60–80 °C). • Degradation: <30 µV/h at 2 A/cm² (AEM, AST steady-state); <1%/500 h performance loss under dynamic cycling protocols. • Measurement resolution: current ±0.1%, voltage ±1 mV, temperature ±0.1 °C, flow ±1%, pressure ±0.1 bar. • Safety: H_2/O_2 cross-leak detection, automated purge/vent, interlocks and SIL-appropriate logic verified by FMEA. • Uptime/throughput: ≥90% availability; ≥5 parallel single-cell channels or ≥2 short-stack channels.			
R&D Plan	WP1 – Requirements & design (M1–M3): Define user and partner specs; interface standards. Milestone: System design freeze. WP2 – Hardware build (M2–M8): assemble fluidic/gas manifolds; integrate power electronics (≤500 A channels); pressure/temperature control. Milestone: Single-cell bench commissioning. WP3 – Diagnostics & software (M3–M10): Implement EIS/HFR, control loops, automated protocols, data model, dashboards. Milestone: Verified measurement accuracy. WP4 – AST & durability methods (M6–M10): Develop AEM-specific ASTs (load cycling, start-stop, impurities); define KPIs. Milestone: Approved SOPs. WP5 – Validation with partner cells and (short) stacks (M8–M14): Integrate 1–10 kW cells and (short) stacks, run performance and durability campaigns. Milestone: Pilot-scale validation report. WP6 – Analyse test results from WP5(M9-M24) to further understand degradation and cell-integration challenges as well as develop improvements with South Korean SME's to repeat WP5 activities to investigate the effect of the improvement. WP7 – Analyse performance of test equipment and test protocols (M9-M24) based on the results and testing experience gained from WP1-WP6 and develop improvements with South Korean SME's to repeat WP5 activities to investigate the effect of the improvement.			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The expected R&R of TNO are: - Overall coordination of the project - Providing lab equipment and facilities - Take the lead in designing and developing a modular AEM single-cell and short-stack test bench. *If a South Korean SME would like to take the lead here, TNO would be open for that.* - Developing the functional requirements for the test equipment with input from South Korean SME - Developing the test protocols which have to validate the test equipment with input from South Korean SME to also ensure the data has value for the South Korean SME component/cell/stack developer. - With money from this program buy the materials of the first prototype and build the first prototype. *If a South Korean SME would like to do this, TNO would be open for that.* - Execute the validation tests for the test equipment - Analyse the data to further understand degradation and cell-integration challenges and bring this back to the partners while taking sensitive IP information of the partners into account. - Reporting and Dissemination of the project
Expected R&R of Korean SME Partner	Multiple Korean SME's are welcome to join this project. This could be interesting for an AEM technology developer and/or a test facility. The expected R&R of the Korean SME Partner(s) is stated below • Supply AEM components, cells, stacks (1–10 kW) which they would like to have tested • Participate in discussions on test system design and test protocol design • Participate in validation testing, failure analysis, and design iterations • Provide feedback on reporting and end-results • Co-develop exploitation and commercialization pathway for Korea/EU markets.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	• Demonstrated AEM test platforms with validated performance envelope and safety case. • Enable high quality AEM validation testing to provide potential customers of the AEM technology with the information they need to invest. • Knowledge transfer between NL and South-Korea on testing and validating AEM electrolyzer technologies. • Enhancing test infrastructure in both NL and South-Korea • Enabling reproducible benchmarking across partners • Degradation insights for the South-Korean SME which can be used to inform potential buyers in materials selection and stack design. • Strengthened EU–Korea supply chain links for electrolyzer testing and component validation.

6. References

(1) Faraday lab: optimising and scaling up electrolysis

(2) News | Accelerating the development of Anion Exchange Membrane electrolysers - VoltaChem

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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